

Part A Chapter 12

Mind Extended in the World¹

1 Introduction

Many animals do not only use their internal cognitive structures to support future behaviour. Often also the external environment is used. A famous example is the fairy tale of how ‘Little Thumb’ dropped a trace of pebbles where he walked so that he was able to find back his way home. Other occurrences of this pattern of behaviour are the use of ‘to do lists’ and ‘lists of desiderata’. Having these written down externally (e.g., on paper, in your diary, in your organizer or computer) makes it unnecessary to have an internal memory about all the items. The only thing to remember is where these lists are available. The states discussed here can be considered *external cognitive states*, as they are used in a way similar to how internal cognitive states are used. In (Clark, 1997, 1999; Clark and Chalmers, 1998) this pattern is discussed in more detail. In this chapter some of these types of behaviour exploiting (manipulated) properties of the external world are analysed. The examples mentioned are examples showing the importance for the study of cognition of taking into account an intensive interaction of the individual with the world.

1.1 Domain description and focus

Clark and Chalmers (1998) point at the similarity between cognitive processes in the head and some processes involving the external world. This similarity can be used as an indication that these processes can be considered extended cognitive processes or extended mind:

‘If, as we confront some task, a part of the world functions as a process which, *were it done in the head*, we would have no hesitation in recognizing as part of the cognitive process, then that part of the world *is* (so we claim) part of the cognitive process. Cognitive processes ain't (all) in the head! (...) Of course, one could always try to explain my action in terms of internal processes and a long series of "inputs" and "actions", but this explanation would be needlessly complex. If an



¹ Based on: Bosse, T., Jonker, C.M., Schut, M.C., and Treur, J., (2005). Simulation and Analysis of Shared Extended Mind. *Simulation Journal (Transactions of the Society for Modelling and Simulation)*, vol. 81, 2005, pp. 719 - 732.

isomorphic process were going on in the head, we would feel no urge to characterize it in this cumbersome way.'

(Clark and Chalmers, 1998)

Behaviour based on extended mind has the following aspects:

- the environment somehow participates in the individual's cognitive processes
- cognitive processes within the individual are simplified
- interaction with the external world becomes more intensive
- the individual has to rely on the external world: delegation of some of its cognitive representations and capabilities (if Little Thumb's pebbles are lost, he is lost; if your to do list is lost, you are lost!)

The story of Little Thumb makes clear how important persistence is, also in this case of external cognitive states. The first time, when he used pebbles, it was silently assumed that pebbles would persist in being where they are. The second time, however, he had no pebbles available but used pieces of bread instead. These did not persist, since animals came to eat them. So, therefore he had to look for another solution that time. A similar invention is used by ants. An ant, for example heading back home from a food source, regularly drops a pheromone on the ground, so that the ant itself, but also its colleagues, can find the way back. So, the fairy tale Little Thumb is a useful metaphor to analyse phenomena in animal (and human) behaviour.

1.2 Characteristic patterns to be explored

From an external behavioural perspective, the behaviour for Little Thumb is quite simple: drop a pebble at each choice point, when he is facing a crossing ahead, and on the way back just observe the trace of pebbles and follow them. Both behaviours are instances of stimulus-response behaviour. Disturbances in the pattern can occur when persistence in the external world is not maintained, for example, when pebbles are taken away by other persons or animals.

2 Analysis of the Main Aspects and their Relations

In this section the main concepts and their relations needed for the model are identified.

2.1 Identification of relevant concepts

As interaction with the environment plays an important role, the concepts of *observation* (of a *crossing*, or a *pebble* at a certain *position*) and *action performance* (*put a pebble*, *go to the position* of a *pebble*) are used. Finally, a *pebble at a certain position* in the world is one of the concepts used. In summary, the following concepts are used:

- observation (of a crossing ahead or a pebble at a certain position)

- action performance (drop a pebble, go to a position, change direction)
- world fact (of position of pebble, piece of bread, self, animal)

2.2 Global structure of the model

Two main parts of the process can be considered: behaviour deciding where to put the pebbles and deciding where to go by observing the pebbles, and the external world process; see Figure 1. Both processes interact. As it is assumed that the behavioural processes are simple, no internal cognitive processes are taken into account.

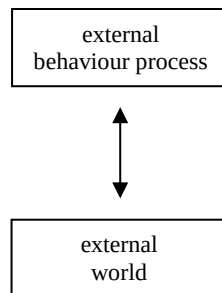


Figure 1 Overall structure of the model

2.3 Relationships between the identified concepts

The dynamic relationships between the identified concepts are as follows; see also Figure 2.

- observation of a position affects dropping a pebble
- observation of a pebble at a certain position affects going to that position
- putting a pebble affects a pebble being at that position
- a pebble being at a position affects observation of the pebble at that position

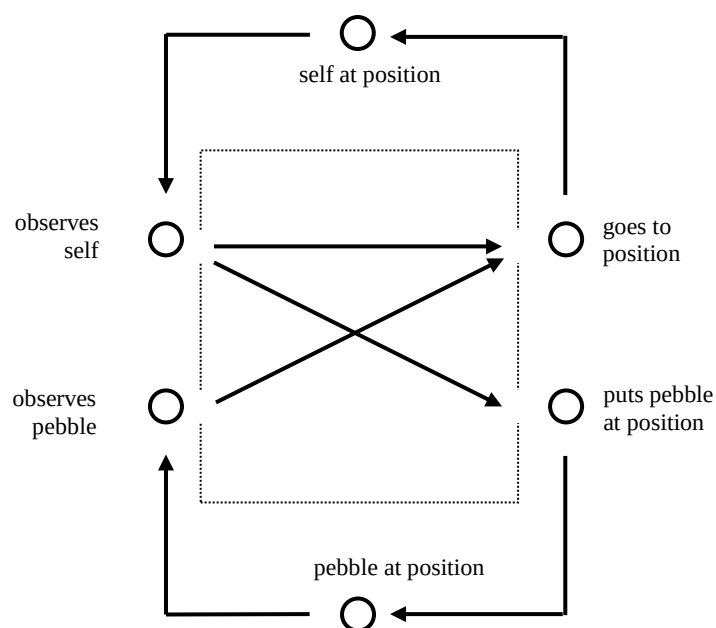


Figure 2 Dynamic relationships between the relevant concepts

3 The Detailed Model

In this section the detailed model is introduced. It is a variation of the model described in (Bosse, Jonker, Schut, and Treur, 2005). First it is shown in which way the relevant concepts are formalised, and next the dynamic relations are formalised in a detailed manner.

3.1 Formalising the concepts used

To obtain a detailed model, first the concepts used are formalised. In this case for each of the concepts a logical atom is introduced. These atoms are based on predicates for observation, performance, sensory representation and belief.

observation (of a position of self or pebble)	observed(at_position(self, P)) observed(at_position(pebble, P))
action performance (drop pebble, go to position, change direction)	performed(drop_pebble) performed(goto(P)) performed(turn_direction)
pebble at a position in external world self at a position in external world adjacent positions in external world direction forward direction backward	world_fact(at_position(pebble, P)) world_fact(at_position(self, P)) world_fact(adjacent(P1, P2)) world_fact(direction_forward) world_fact(direction_backward)

Table 1 Formalisations of the relevant concepts by logical atoms

3.2 Formalising the relationships

The dynamics of these concepts involve temporal relationships on the behaviour of the individual, and of the world, which are as follows.

LP1 Dropping a pebble

at any point in time
 if the individual observes its position P
 and the direction is forward
 and there is no pebble at this position P
 and there was no pebble dropping action yet
 and P is not p3
 then it will drop a pebble

Formally:

$$\text{observed}(\text{at_position}(\text{self}, P)) \wedge \text{world_fact}(\text{direction_forward}) \wedge \text{not world_fact}(\text{at_position}(\text{pebble}, P)) \wedge \text{not performed}(\text{drop_pebble}) \wedge P \neq p3 \rightarrow \text{performed}(\text{drop_pebble})$$

LP2 Moving forward

at any point in time
 if the individual observes its position P1
 and it observes that P2 is an adjacent position
 and the direction is forward
 and it did not go to P2 yet
 and it did not arrive at P2 yet
 then the individual will go to P2

Formally:

$$\text{observed}(\text{at_position}(\text{self}, P1)) \wedge \text{observed}(\text{adjacent}(P1, P2)) \wedge \text{world_fact}(\text{direction_forward}) \wedge \text{not performed}(\text{goto}(P2)) \\ \wedge \text{not world_fact}(\text{at_position}(\text{self}, P2)) \rightarrow \text{performed}(\text{goto}(P2))$$
LP3 Turning around

at any point in time
 if the individual observes its position p3
 and the direction is not backward
 and it did not just turn direction
 then the individual will turn direction

Formally:

$$\text{observed}(\text{at_position}(\text{self}, p3)) \wedge \text{not world_fact}(\text{direction_backward}) \wedge \text{not performed}(\text{turn_direction}) \rightarrow \text{performed}(\text{turn_direction})$$
LP4 Moving backward to pebbles

at any point in time
 if the individual observes a pebble at a position P2
 and the direction is backward
 and it observes its own position P1
 and it observes that P2 is an adjacent position
 and it did not go to P2 yet
 then the individual will go in the direction of that position

Formally:

$$\text{observed}(\text{at_position}(\text{pebble}, P2)) \wedge \text{world_fact}(\text{direction_backward}) \wedge \text{observed}(\text{adjacent}(P2, P1)) \wedge \\ \text{observed}(\text{at_position}(\text{self}, P1)) \wedge \text{not performed}(\text{goto}(P2)) \rightarrow \text{performed}(\text{goto}(P2))$$

Notice that this relationship describes just stimulus-response behaviour. In fact the behaviour as such is a kind of simulation of delayed response behaviour by means of stimulus-response behaviour, in such a manner that no internal memory whatsoever is needed. In fact the external world serves as a form of external memory. This shows the strength of this type of behaviour. No persisting internal mental structures are required. To be able to use external memory, the creation and persistence of the external memory has to be arranged as well.

EP1 Action effect for pebble dropping

at any point in time

if the individual is at a certain position
 and a pebble is dropped
 then this pebble will be at this position

Formally:

$$\text{world_fact}(\text{at_position}(\text{self}, P)) \wedge \text{performed}(\text{drop_pebble}) \rightarrow \text{world_fact}(\text{at_position}(\text{pebble}, P))$$

EP2 Persistence of pebbles

at any point in time
 if a pebble is at a certain position
 then at all later time points this pebble will be at this position

Formally:

$$\text{world_fact}(\text{at_position}(\text{pebble}, P)) \rightarrow \text{world_fact}(\text{at_position}(\text{pebble}, P))$$

EP3 Action effect for going to a position

at any point in time
 if the individual goes to a certain position
 then he will be at this position

Formally:

$$\text{performed}(\text{goto}(P)) \rightarrow \text{world_fact}(\text{at_position}(\text{self}, P))$$

EP4 Action effect for turning around

at any point in time
 if the individual turns direction
 then the direction will be backward

Formally:

$$\text{performed}(\text{turn_direction}) \rightarrow \text{world_fact}(\text{direction_backward})$$

EP5 Persistence of forward direction

at any point in time
 if the individual has forward direction
 and he is not at position p3
 then he will have this direction

Formally:

$$\text{world_fact}(\text{direction_forward}) \wedge \text{not world_fact}(\text{at_position}(\text{self}, p3)) \rightarrow \text{world_fact}(\text{direction_forward})$$

EP6 Persistence of backward direction

at any point in time
 if the individual has backward direction
 then he will have this direction

Formally:

$$\text{world_fact}(\text{direction_backward}) \rightarrow \text{world_fact}(\text{direction_backward})$$

EP7 Persistence of being at a position

at any point in time
 if the individual is at a position P1

and for all positions P2 he is not going to P2
 then he will be in position P1

Formally:

$$\text{world_fact}(\text{at_position}(\text{self}, P1)) \wedge [\forall P2:\text{Positions} \text{ not performed}(\text{goto}(P2))] \rightarrow \text{world_fact}(\text{at_position}(\text{self}, P1))$$

Note that the expression $\forall P2:\text{Positions} \text{ not performed}(\text{goto}(P2))$ is meant is a short notation for the conjunction of all literals $\text{not performed}(\text{goto}(P2))$ with for P2 all specific position names substituted:

$$\text{not performed}(\text{goto}(p1)) \wedge \text{not performed}(\text{goto}(p2)) \wedge \text{not performed}(\text{goto}(p3))$$

EP8 Observation of pebble at own position

at any point in time
 if the individual is at a certain position
 and a pebble is present at this position
 then he will observe the pebble at this position

Formally:

$$\text{world_fact}(\text{at_position}(\text{self}, P)) \wedge \text{world_fact}(\text{at_position}(\text{pebble}, P)) \rightarrow \text{observed}(\text{at_position}(\text{pebble}, P))$$

EP9 Observation of pebble at adjacent position

at any point in time
 if the individual is at a certain position P1
 and a pebble is present at adjacent position P2
 and the direction is backward
 then he will observe the pebble at this position

Formally:

$$\text{world_fact}(\text{at_position}(\text{self}, P1)) \wedge \text{world_fact}(\text{adjacent}(P2, P1)) \wedge \text{world_fact}(\text{at_position}(\text{pebble}, P2)) \wedge \text{world_fact}(\text{direction_backward}) \rightarrow \text{observed}(\text{at_position}(\text{pebble}, P2))$$

EP10 Observation of self

at any point in time
 if the individual is at a certain position P
 then he will observe that he is at this position

Formally:

$$\text{world_fact}(\text{at_position}(\text{self}, P)) \rightarrow \text{observed}(\text{at_position}(\text{self}, P))$$

EP11 Observation of adjacent positions

at any point in time
 if positions P1 and P2 are adjacent
 then the individual will observe this

Formally:

$$\text{world_fact}(\text{adjacent}(P1, P2)) \rightarrow \text{observed}(\text{adjacent}(P1, P2))$$

From these relationships only the relationships LP1 to LP4 directly relates to the individual. The relationships EP1 to EP11 are on the external world only. Notice the similarity of these external dynamic relationships to the internal dynamic relationships in the case of delayed response behaviour as described in Part B, Chapter 2.

In the second part of the story Little Thumb uses pieces of bread instead of pebbles. This may cause (and actually did cause) serious problems, as pieces of bread are less persistent than pebbles: they can be eaten by animals. Then the above persistence relationship **EP8** of the world has to be replaced by:

EP12 Conditional persistence of bread pieces

at any point in time
 if a piece of bread is at a certain position
 and no animal is present at this position
 then this piece of bread will be present at this position

Formally:

$$\text{world_fact(at_position(bread, P))} \wedge \text{not world_fact(at_position(animal, P))} \rightarrow \text{world_fact(at_position(bread, P))}$$

Notice that here the more complex structures are found in the outside world instead of in the internal part, as was the case in previous chapters. In contrast, here the internal structures are limited to stimulus-response associations. This illustrates as a small example, what in (Clark, 1999) is claimed: the more we involve our external world in the cognitive processes, the more intelligent can our behaviour be, without complex internal cognitive processing.

4 Simulation Experiments

Based on the dynamic relationships discussed in the previous section, simulations have been performed in the LEADSTO environment. In Figure 3 an example simulation trace is shown. It is shown how until time point 10 the forward moving takes place while dropping pebbles. After that the direction is changed and the pebbles are observed and followed.

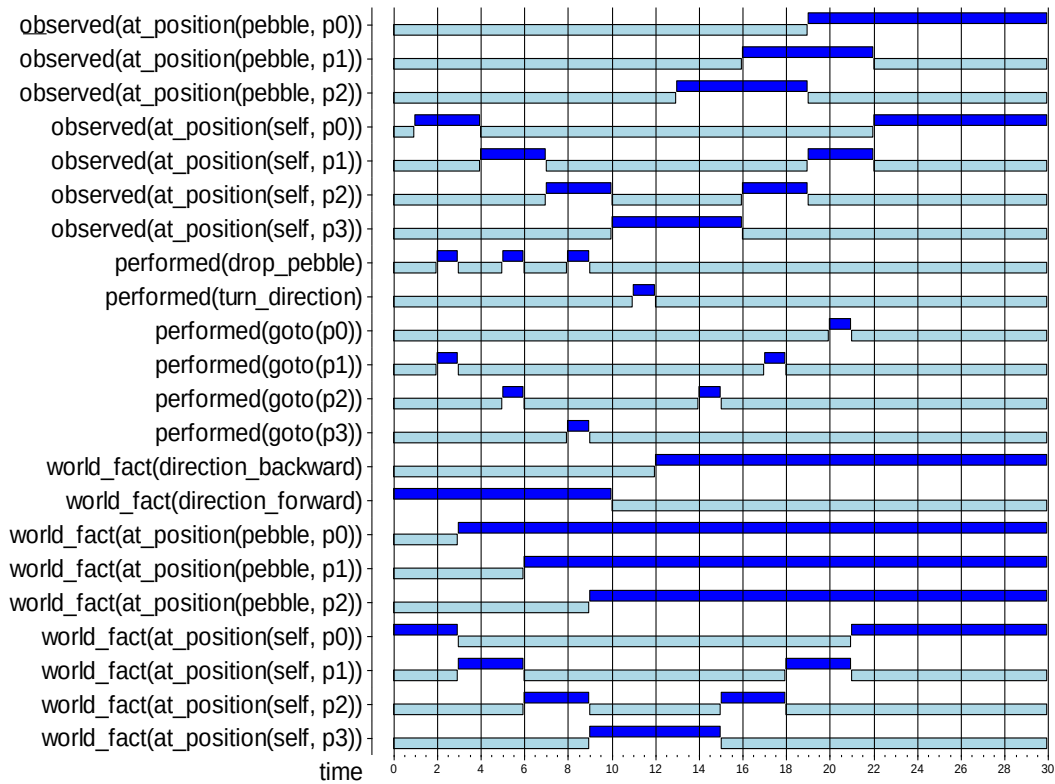


Figure 3 Example simulation trace

5 Evaluation

In this section, the simulations performed using the model are evaluated with respect to the characteristic patterns identified in Section 1.2. Moreover, the assumptions behind the model are summarised. Finally some possibilities for variations or refinements of the model are discussed.

5.1 Evaluation of the simulations and characteristic patterns

A number of different simulations conducted showed the characteristic patterns identified in Section 1.2. It is easy to perform a few more experiments to check more scenarios.

5.2 Assumptions made

The following assumptions were made.

- Pebbles and their positions persist.
- No other pebbles are there than the ones put by Little Thumb.
- Pebbles are always found back by observation.
- The way back is clearly distinguished from the way forth. For example, on the way forth no pebbles are followed, and on the way back no pebbles are dropped.

5.3 Possible variations and refinements

Variants and refinements of the model can be made by taking into account the following possibilities.

- Take bread pieces instead of pebbles, with less persistence. How does it work when other animals play with them and then change their positions, or take them away.

6 Further Explorations

A number of further issues can be addressed for the analysis of behaviour based on extended mind.

- How can an extended mind be shared by multiple individuals? For example, pebbles are dropped so that others can follow the same route.
- How can motivation-based behaviour be modelled by stimulus-response behaviour, making use of an external ‘list of desiderata’ instead of internal desires, and an external ‘to do list’ instead of internal intentions?
- Another type of phenomena where extended mind plays a crucial role is in the organisation of (multiple) ants behaviour. While they walk, ants drop pheromones on the ground. Other ants sense these pheromones and follow the route in the direction of the strongest sensing. Pheromones are not persistent for long times; they decay. Therefore such routes can vary over time. Although the basic principles have a strong similarity to those in the Little Thumb case, a difference is that not only the individual itself uses the sensing of the pheromones but also other society members.

References

- Bosse, T., Jonker, C.M., Schut, M.C., and Treur, J., (2005). Simulation and Analysis of Shared Extended Mind. *Simulation Journal (Transactions of the Society for Modelling and Simulation)*, vol. 81, 2005, pp. 719 - 732.
- Clark, A. (1997). *Being There: Putting Brain, Body and World Together Again*. MIT Press, 1997.
- Clark, A. (1999). Where brain, body, and world collide. *Journal of Cognitive Systems Research*, 1, 1999, pp. 5-17.
- Clark, A., and Chalmers, D. (1998). The Extended Mind. In: *Analysis*, vol. 58, pp. 7-19.